

Body Science Checkpoint

BUILD • Science • 40 minutes

I can use models and evidence to show what I know about skeletons, muscle pairs, nutrition, balance and food sources.

Prepare before pupils arrive

Setup: Set four low-clutter stations and a complete card-only route. Use the live W2–W7 taught content, not stale RAG outcomes. Decide in advance how each strand will trigger re-model, contrast, transfer or access change.

Safety: Check models and small parts, maintain no-touch and non-speaking routes, use no real food and protect pupils from personal body/diet questions, public scoring or surprise assessment.

Equipment: Skeleton or labelled diagram; vertebrate/invertebrate cards; safe card-and-split-pin arm model; nutrition/body-job cards; food-group mats; simple food-chain cards; tiered checkpoint sheet.

Safety: Teach bodies and food on models, diagrams and supplied cards, never by judging a pupil's body, diet or home. No food is eaten. Check split pins and small parts, offer a no-touch route and avoid public scores or rank order.

Fallback: A pupil may point, sign, operate one switch, direct an adult, use eye gaze or have exact words scribed. Keep the science demand while changing access.

Access and participation

Offer reading aloud, pointing, signing, quiet thinking, a no-touch route and exact scribing where these support the learner. Keep the same subject decision. In shared work, let each learner commit before feedback; use a partner as card mover or checker, then swap. Avoid public speed rankings.

Source lesson curriculum link: Sugar investigation and Autumn 1 consolidation: skeletons, muscles and nutrition.

40-minute teaching sequence

Stage / total time	Teaching move	Adult support / response
Arrival · 3 min	Name one skeleton job and one bone example. Explain what one muscle does while its partner relaxes.	Wait, regulate and retrieve through the lightest workable route. If recall is inaccessible, point to one retrieval sentence; do not teach today's answer.
Starter · 3 min	Think first. Point, say, sign or write your choice.	Protect curiosity: receive every first idea without judgement. Ask 'What makes you think that?' and preserve the prediction for later evidence.

Stage / total time	Teaching move	Adult support / response
I do · 4 min	support — hold up and give shape contract — shorten and pull nutrition — getting what the body needs from food Checkpoint 1: use a skeleton model to match support, protection and movement to an example; classify one animal using backbone evidence. Checkpoint 2: operate the arm model.	Let the teacher/model carry the explanation; pupils watch, point or track. Use a visual cue only if access is the barrier, then fade it.
We do · 4 min	Commit to an answer. Show the evidence you used.	Scan every response mode. Do not infer understanding from handwriting or confidence. Secure: transfer. Mixed: contrast. Misconception: counterexample then a fresh question.
I do 2 · 3 min	Checkpoint 3: match food evidence to body jobs such as energy, growth and repair; explain why variety matters. Checkpoint 4: sort foods with a stated rule and build a simple producer-to-consumer food-source chain.	Model one complete evidence-to-explanation sentence, then pause. Keep the science words; change density, modality or adult role before changing the goal.
We do 2 · 4 min	Route each evidence card to the strand it checks. Then choose one strand to explain and one to revisit.	Protect pupil operation and thinking; support handling only where needed. Ask what changed and what stayed the same before supplying a scientific word.
Independent · 15 min	Complete or direct four short checkpoint stations, preserve a piece of evidence from each and explain one connection between two strands. Record: One observable response from each strand, one because connection, the pupil's chosen next strand and an adult witness note naming independence and prompts.	Protect independence. Accept equivalent evidence routes and scribe exact meaning. Prompt only as much as needed; fade as soon as the pupil continues.
Exit · 4 min	Explain one connection between skeletons, muscles, nutrition or food sources using evidence. Point to, read or explain the evidence you made.	Genuinely receive one final response and name back the Science heard or seen. Choose one visible next move; written closure is optional when Voice and Audience are observable.

The timing belongs to the whole stage. Several PowerPoint slides may share it; do not restart that time on every slide. Full source-stage text is in the PowerPoint speaker notes and original HTML.

Answers, evidence and feedback

Hinge answer: Relax while the biceps pulls the forearm bone.

Yes. In the simple pair model, the biceps contracts and pulls while the triceps relaxes.

Misconception repair

Repaired. Movement links skeletons, joints and muscle pairs; nutrition claims must stay within the evidence.

Guided checkpoint

Yes. In the simple pair model, the biceps contracts and pulls while the triceps relaxes. Muscles can pull when they contract. Which partner must lengthen or relax for the movement?

Skull around the brain

SKELETON / BACKBONE. Skeleton strand: the skull protects the brain. Match this named bone to a skeleton job.

Snake X-ray shows an internal spine

SKELETON / BACKBONE. Skeleton strand: backbone evidence classifies the snake as a vertebrate. Legs do not decide this group; look for the internal backbone.

Biceps becomes shorter

MUSCLE PAIR. Muscle strand: contraction produces a pull on the bone. Place the card with the system that contracts and pulls.

Partner relaxes during the bend

MUSCLE PAIR. Muscle strand: the pair changes roles for movement in the opposite direction. This card describes the partner in an opposing muscle pair.

Energy, growth and repair

NUTRITION / BALANCE. Nutrition strand: food provides nutrients used for several body jobs. Match these words to what the body obtains from food.

A varied pattern across food groups

NUTRITION / BALANCE. Nutrition strand: balance uses a range rather than one perfect item. This card concerns variety and balance, not where an animal's food began.

Plant → caterpillar

FOOD SOURCE / CHAIN. Food-source strand: the caterpillar obtains food by eating the plant. Follow the arrow from food to feeder.

Caterpillar → bird

FOOD SOURCE / CHAIN. Food-source strand: the bird obtains food from the caterpillar. This card traces where an animal gets food.

Model limitation

Models simplify bodies, joints, diets and food chains. This checkpoint samples key ideas; it is not a medical, dietary or full attainment diagnosis.

Independent evidence

One observable response from each strand, one because connection, the pupil's chosen next strand and an adult witness note naming independence and prompts.

Supported: Complete one two-choice item at each station, then choose a secure strand and a next strand. Use symbols, model handling or pointing. Read each option exactly and scribe the pupil's own explanation.

Standard: Complete all four stations, justify one response from each and connect two strands using because. Use: ___ works with ___ because ___. My evidence is ___. Next I need to practise ___.

Stretch: Explain two cross-strand links, repair one misconception and evaluate one model limitation. Keep the claim within the model and distinguish evidence from a wider health judgement.

Record the teaching response

What the learner actually showed	Next teaching action
Secure explanation with evidence	Reduce content prompting; offer another example or a limitation.
Partly correct / mixed	Point to the deciding evidence and invite a revision.
Access barrier	Change the reading, sensory or response format; keep the subject decision.
Missing source or observation	Keep the gap visible. Name the source or authorised check needed.

Safety and evidence boundary

Use the centre's authorised practical or fieldwork method and local risk assessment. Supplied sample data, fictional place models, card feedback and rehearsals are teaching material. They are not the learner's practical observations or proof of a qualification outcome. For portfolio use, check the exact registered unit/version, accepted evidence and centre process.

Sources and companion notes

Primary classroom source: SCI_B_W8B_Autumn_Science_Checkpoint_Do.html

The uploaded lesson is included unchanged. Text and teaching steps in the PowerPoint are editable; source diagrams are placed images. Word booklets are editable. Static PDFs cannot reproduce HTML controls; use the paper cards/tables or open the original HTML.

Verification scope: matched to the uploaded title, pathway, objective, stage sequence and 40-minute timing. Embedded workbook references are retained as source locators; this is not a new line-by-line audit of every scheme-of-work row or a centre assessment sign-off.